620 Web analytics final project proposal:

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Project objective:

In this project, we will be analyzing social media (Twitter, Facebook, and Instagram) network and post-sentiments between users. Specifically, the following three aspects will be examined.

1. Do users who share the same hashtag on social media exhibit similar sentiment patterns?
2. Can we predict post sentiment from both textual content and social network positions?
3. Can social network position predict user sentiment?

From a business perspective, understanding how sentiments are formed and what structural factors influence them has direct commercial value. If users within the same hashtag community tend to share similar sentiment patterns, advertisers can move beyond broad demographic targeting and instead place campaigns within communities that are already predisposed toward a particular emotional tone — matching product messaging to sentiment context rather than just topic. More broadly, the ability to predict user sentiment from network position alone, without relying on text content, opens the door to real-time sentiment targeting in environments where post content is sparse or unavailable.

Dataset Source: https://www.kaggle.com/datasets/kashishparmar02/social-media-sentiments-analysis-dataset

Preview of the dataset:

Text	Sentiment	Timestamp	User	Platform	Hashtags	Retweets	Likes	Country	Year	Month	Day	Hour
Enjoying a beautiful day at the park! ...	Positive	2023-01-15 12:30:00	User123	Twitter	#Nature #Park	15.0	30.0	USA	2023	1	15	12
Traffic was terrible this morning. ...	Negative	2023-01-15 08:45:00	CommuterX	Twitter	#Traffic #Morning	5.0	10.0	Canada	2023	1	15	8
Just finished an amazing workout! 💪 ...	Positive	2023-01-15 15:45:00	FitnessFan	Instagram	#Fitness #Workout	20.0	40.0	USA	2023	1	15	15
Excited about the upcoming weekend getaway! ...	Positive	2023-01-15 18:20:00	AdventureX	Facebook	#Travel #Adventure	8.0	15.0	UK	2023	1	15	18
Trying out a new recipe for dinner tonight. ...	Neutral	2023-01-15 19:55:00	ChefCook	Instagram	#Cooking #Food	12.0	25.0	Australia	2023	1	15	19

Methods:

Before answering the first question, the "Sentiment" column has to be condensed since the original dataset contained over 200 unique sentiment labels. The VADER library, specialized in social media semantics, was used to analyze the text sentiments and the polarity score of the text was used to categorize the overall sentiment into three groups: positive, negative, and neutral. The VADER labels were then compared to the original "Sentiment" column, which is also complied by the VADER library into the same three categories, for agreement rate comparison. We will define an agreement rate > 70% as acceptable for the VADER labels to replace the original target variable due to its diversity in sentiments.

<u>Q1: Sentiment comparison between communties (NLP/Text processing)</u>

Once the new target variable is established, we will employ Chi-Square test to statistically determine if different network communities share significantly different sentiments. For example, is the overall sentiment from community number 1, which shared the hashtag "Nature", different from the community number 2, which shared the hashtag "Food". The null hypothesis is that different communities do not show different sentiment in their posts. The alternative hypothesis is that different communities have significant different sentiment in their posts.

Q2: Sentiment prediction based on NLP/Text processing and social network analysis (post-level):

To answer the second question, we will extract text and network features such as degree centralities, pos-neighbor-ratio, and betweenness and input all these features into 3 different machine-learning (ML) algorithms Logistic Regression, RandomForest, and Stochastic Gradient Descent (SGD) to classify and compare the results with train test split.

The main evaluation metric is the average F1 macro score since the three classes in the target variable are imbalanced, where the positive sentiment constitutes the majority class. Therefore, the average F1 macro score is a better evaluator than the accuracy score, which can falsely elevate the metric by focusing only on the scores of the major class and ignoring minority is poorly classified.

A baseline metrics generated with text-features only will be compared against metrics by text features and social network features. A direct average F1 macro score will show if post-level sentiment can be predicted by text features alone or both text and network features will further enhance the prediction.

Q3: Sentiment prediction based solely on user-level social networks (user-level):

Q3 is fundamentally different from Q2 because it looks at the user-level, removing the post-level(texts) entirely. Features such as average likes, single post only, clustering, degree of connection, social media platforms usage are extracted to use for training ML models for sentiment predictions. The result of this section will answer the question whether users can be influenced by his or her neighbors to share similar sentiment. For instance, if a person is linked to a group of food-oriented community, will him or her be also sentimentally express similar semantics as this community? The metrics used will also be average F1 macro score due to the same class imbalance issue mentioned above.

Summary:

This study covers the intersection of social network analysis and natural language processing applied to multi-platform social media data. By constructing a user-user network based on shared hashtags and combining it with VADER-derived sentiment labels, we examine whether community membership, text content, and network position can collectively explain and predict how users express sentiment online. The three questions progress in scope — from statistical association (Q1), to post-level prediction combining text and network features (Q2), to user-level prediction using network structure alone (Q3) — allowing us to isolate the independent contribution of network position to sentiment behavior. If network features alone approach the predictive power of text features, it would suggest that a user's social context is as informative as what they actually write, with broader implications for understanding echo chambers and sentiment homophily in online communities.

Limitations:

Synthetic data: The dataset is suspected to be synthetic rather than scraped comments. The degree of effect is yet to be determined.

Reproducibility: The Louvain algorithm is non-deterministic — community assignments may differ across runs, which affects Q1 and Q3 results.

Network sparsity: If many users share few hashtags, the graph may be too sparse for community detection to be meaningful.

VADER threshold: The 70% agreement threshold is defined based on the circumstantial factors such as high number of unique labels.